Session 14B: Advanced Differentiation — Product, Quotient, Chain, and Implicit

Phase 2 — Classical Techniques | 80 min

Simple functions differentiate with a dictionary lookup. Products, quotients, and nested functions need rules. This session gives you a decision tree: identify the structure, apply the matching procedure. By the end, you can differentiate anything built from elementary functions.

Prerequisites: 14A (basic derivative dictionary), 10A (exponents & logs), 11A (trig)

The Differentiation Decision Tree

> **Before you start**, identify the outermost structure. Match it to a procedure.

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Look at the function. What is the LAST operation you would perform when evaluating?

1. Is it f ± g?                     → Split and differentiate each piece (14A)
2. Is it f × g (product)?           → PRODUCT RULE (14B-A)
3. Is it f ÷ g (quotient)?          → QUOTIENT RULE (14B-A)
4. Is it f(g(x)) (composition)?     → CHAIN RULE (14B-B)
5. Are x and y mixed together?      → IMPLICIT DIFFERENTIATION (14B-C)
6. Is it f(x)^{g(x)}?               → LOGARITHMIC DIFFERENTIATION (14B-C)
7. Is it parametric (x(t),y(t))? → PARAMETRIC: dy/dx = (dy/dt)/(dx/dt) (14B-D)
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Part A: Product and Quotient — Multiplication and Division

Example 1: Product Rule — $(fg)' = f'g + fg'$

Procedure:

Step	Action
1	Identify the **two factors**: $f(x)$ and $g(x)$
2	Differentiate the **first**: $f'(x)$. Leave the second alone.
3	Differentiate the **second**: $g'(x)$. Leave the first alone.
4	**Add** the two products: $f'(x)g(x) + f(x)g'(x)$
5	Simplify — factor common terms if possible

Memory: "First derivative × second + first × second derivative."

$\frac{d}{dx}(x^2 \sin x)$:

Step	Action
1	$f = x^2$, $g = \sin x$
2	$f' = 2x$
3	$g' = \cos x$
4	$2x \cdot \sin x + x^2 \cdot \cos x = 2x \sin x + x^2 \cos x$

Triple product: $(fgh)' = f'gh + fg'h + fgh'$. Pattern: differentiate one factor at a time.

$\frac{d}{dx}(xe^x \sin x) = 1 \cdot e^x \sin x + x \cdot e^x \sin x + xe^x \cdot \cos x = e^x(\sin x + x \sin x + x \cos x)$.

Example 2: Quotient Rule — $(f/g)' = \frac{f'g - fg'}{g^2}$

Procedure:

Step	Action
1	Identify **top** $f(x)$ and **bottom** $g(x)$
2	Differentiate top: $f'(x)$. Differentiate bottom: $g'(x)$
3	Assemble numerator: $f'(x)g(x) - f(x)g'(x)$ (top derivative FIRST)
4	Denominator: $[g(x)]^2$
5	Simplify algebraically — expand, cancel, factor

Memory: "Low d-high minus high d-low, over low squared."

$\frac{d}{dx}\frac{x^2}{x+1}$:

Step	Action
1	$f = x^2$ (top), $g = x + 1$ (bottom)
2	$f' = 2x$, $g' = 1$
3	Numerator: $2x(x+1) - x^2(1) = 2x^2 + 2x - x^2 = x^2 + 2x$
4	Denominator: $(x+1)^2$
5	$\frac{x^2+2x}{(x+1)^2} = \frac{x(x+2)}{(x+1)^2}$

Example 3: Product vs. Quotient — The Decision

If...	Use...	Because...
Two factors multiplied	Product rule	$(fg)' = f'g + fg'$

If...	Use...	Because...
A fraction	Quotient rule	$(f/g)' = (f'g - fg')/g^2$
A fraction where bottom is simple (x^n)	Rewrite as $f \cdot x^{-n}$, use product rule	Often faster — avoids quotient rule algebra

$\frac{d}{dx}\frac{\sin x}{x}$: Use quotient rule. $f = \sin x$, $g = x$. $\frac{\cos x \cdot x - \sin x \cdot 1}{x^2} = \frac{x \cos x - \sin x}{x^2}$.

$\frac{d}{dx}\frac{3}{x^2}$: Rewrite as $3x^{-2}$. Power rule: $-6x^{-3} = -\frac{6}{x^3}$. Faster than quotient rule.

Part B: The Chain Rule — Differentiating Nested Functions

> **The most important rule in all of calculus.** When one function is inside another, differentiate layer by layer from outside in.

Example 4: The Chain Rule Algorithm — Peel the Onion

Procedure for $(f \circ g)(x) = f(g(x))$:

Step	Action
1	Identify the **outer function** f (the last thing you'd evaluate)
2	Identify the **inner function** g (inside the parentheses, root, exponent, etc.)
3	Differentiate the **outer**: $f'(\text{leave inside untouched})$
4	Multiply by the derivative of the **inner**: $g'(x)$

Memory: "Derivative of outside × derivative of inside."

Example 5: Basic Chain Rule — One Layer

$\frac{d}{dx}(x^2+1)^5$:

Step	Action
1	Outer: $(\square)^5 \to$ power rule on the outside
2	Inner: $\square = x^2+1$
3	Outer derivative: $5(x^2+1)^4$ (bring 5 down, leave inside untouched)
4	Inner derivative: $2x$. Multiply: $5(x^2+1)^4 \cdot 2x = 10x(x^2+1)^4$

$\frac{d}{dx}\sin(x^3)$: Outer = $\sin(\square) \to \cos(\square)$. Inner = $x^3 \to 3x^2$. Result: $\cos(x^3) \cdot 3x^2 = 3x^2\cos(x^3)$.

$\frac{d}{dx}e^{\sin x}$: Outer = $e^{\square} \to e^{\square}$. Inner = $\sin x \to \cos x$. Result: $e^{\sin x}\cos x$.

$\frac{d}{dx}\ln(\cos x)$: Outer = $\ln(\square) \to 1/\square$. Inner = $\cos x \to -\sin x$. Result: $\frac{1}{\cos x} \cdot (-\sin x) = -\tan x$.

Example 6: Nested Chain — Two or Three Layers

Procedure for multiple layers: Peel from outermost to innermost, writing each derivative in a chain. Multiply them all together.

$\frac{d}{dx}\sin(e^{x^2})$:

Layer	Function	Derivative
Outermost	$\sin(\square)$	$\cos(e^{x^2})$
Middle	$e^{\square}$	e^{x^2}
Innermost	x^2	$2x$

Multiply all three: $\cos(e^{x^2}) \cdot e^{x^2} \cdot 2x = 2xe^{x^2}\cos(e^{x^2})$.

$\frac{d}{dx}\sqrt{\ln(\sin x)}$:

Layer	Derivative
$\sqrt{\square} = (\square)^{1/2}$	$\frac{1}{2}(\ln(\sin x))^{-1/2}$
$\ln(\square)$	$\frac{1}{\sin x}$
$\sin x$	$\cos x$

Multiply: $\frac{1}{2\sqrt{\ln(\sin x)}} \cdot \frac{1}{\sin x} \cdot \cos x = \frac{\cot x}{2\sqrt{\ln(\sin x)}}$.

Example 7: Chain Rule Quick Reference

Outer form	Outer derivative	Example
$(\square)^n$	$n(\square)^{n-1}$	$(x^3+1)^4 \to 4(x^3+1)^3 \cdot 3x^2$
$\sin(\square)$	$\cos(\square)$	$\sin(5x) \to \cos(5x) \cdot 5$
$\cos(\square)$	$-\sin(\square)$	$\cos(x^2) \to -\sin(x^2) \cdot 2x$
$e^{\square}$	$e^{\square}$	$e^{3x} \to e^{3x} \cdot 3$
$\ln(\square)$	$1/\square$	$\ln(x^2+1) \to \frac{1}{x^2+1} \cdot 2x$
$\tan(\square)$	$\sec^2(\square)$	$\tan(x^3) \to \sec^2(x^3) \cdot 3x^2$

Part C: Implicit and Logarithmic Differentiation

Example 8: Implicit Differentiation — When y Is Hidden

Trigger: x and y are mixed together in an equation (not $y =$ something).

Procedure:

Step	Action
1	Differentiate **both sides** with respect to x
2	Whenever you differentiate a y-term, **multiply by** $\frac{dy}{dx}$ (treat y as a function of x)
3	**Collect** all terms with $\frac{dy}{dx}$ on one side, everything else on the other
4	**Factor out** $\frac{dy}{dx}$, then **solve** for it

$x^2 + y^2 = 25$. Find $\frac{dy}{dx}$.

Step	Action
1	$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$
2	$2x + 2y\frac{dy}{dx} = 0$ (the y^2 needs $\frac{dy}{dx}$!)
3	$2y\frac{dy}{dx} = -2x$
4	$\frac{dy}{dx} = -\frac{x}{y}$

Check: At $(3, 4)$ on the circle, slope = $-3/4$. The tangent is perpendicular to the radius (slope $4/3$). Product of slopes = -1 ✓.

Example 9: Implicit with Product — x and y Multiplied

$x^3 + y^3 = 6xy$ at $(3, 3)$. Find $\frac{dy}{dx}$.

1. $\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) = \frac{d}{dx}(6xy)$.
2. $3x^2 + 3y^2\frac{dy}{dx} = 6(y + x\frac{dy}{dx})$. (Right side needs product rule on xy!)
3. $3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$.
4. $3y^2\frac{dy}{dx} - 6x\frac{dy}{dx} = 6y - 3x^2$.
5. $\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2 \to \frac{dy}{dx} = \frac{6y-3x^2}{3y^2-6x}$.
6. At $(3,3)$: $\frac{18-27}{27-18} = -1$.

Example 10: Logarithmic Differentiation — x^x and Friends

Trigger: Both base AND exponent contain x (e.g., x^x, $(\sin x)^{\cos x}$, $(x^2+1)^x$).

Procedure:

Step	Action
1	Set $y = f(x)$. Take $\ln$ of **both sides**: $\ln y = \ln(f(x))$
2	Use log laws to **simplify** the right side: $\ln(a^b) = b\ln a$
3	Differentiate **both sides** with respect to x. Left side: $\frac{1}{y}\frac{dy}{dx}$. Right side: use product/chain as needed
4	**Multiply** both sides by y to isolate $\frac{dy}{dx}$
5	**Replace** y with the original $f(x)$

$y = x^x$:

Step	Action
1,2	$\ln y = x\ln x$
3	$\frac{1}{y}\frac{dy}{dx} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$
4,5	$\frac{dy}{dx} = x^x(\ln x + 1)$

$y = (x^2+1)^{\sin x}$:

1,2. $\ln y = \sin x \cdot \ln(x^2+1)$.
3. $\frac{y'}{y} = \cos x \ln(x^2+1) + \sin x \cdot \frac{2x}{x^2+1}$.
4,5. $y' = (x^2+1)^{\sin x}\left[\cos x \ln(x^2+1) + \frac{2x \sin x}{x^2+1}\right]$.

Part D: Parametric and Inverse Function Derivatives

Example 11: Parametric — $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Trigger: x and y are given as functions of a third variable t.

Procedure:

1. Differentiate $y(t)$ with respect to $t \to dy/dt$.
2. Differentiate $x(t)$ with respect to $t \to dx/dt$.
3. Divide: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

$x = t^2, y = t^3$: $\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$.

$x = \cos t, y = \sin t$ at $t = \pi/4$: $\frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t$. At $t = \pi/4$: -1.

Example 12: Inverse Function Derivative

If $y = f(x)$ and f is invertible: $(f^{-1})'(y) = \frac{1}{f'(x)}$ where $y = f(x)$.

Procedure: To find $(f^{-1})'(b)$:

1. Solve $f(a) = b$ to find a.

2. Compute $f'(a)$.
3. Answer = $1/f'(a)$.

$f(x) = x^3 + x$. Find $(f^{-1})'(2)$.

1. $f(1) = 1^3 + 1 = 2$, so $a = 1$.
2. $f'(x) = 3x^2 + 1$, $f'(1) = 4$.
3. $(f^{-1})'(2) = 1/4$.

Example 13: Inverse Trig Derivatives — Quick Reference

$f(x)$	$f'(x)$	**Domain**
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$\lvert x\rvert < 1$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	$\lvert x\rvert < 1$
$\arctan x$	$\frac{1}{1+x^2}$	all real x

Combined with chain rule: $\frac{d}{dx}\arcsin(2x) = \frac{2}{\sqrt{1-4x^2}}$. $\frac{d}{dx}\arctan(x^2) = \frac{2x}{1+x^4}$.

Example 14: Absolute Value and Piecewise — Check the Break Points

$|x|$: $f'(x) = 1$ for $x > 0$, $f'(x) = -1$ for $x < 0$, **undefined** at $x = 0$ (corner).

$|x^2 - 1|$: Breaks where $x^2 - 1 = 0 \to x = \pm 1$.

- $x < -1$ or $x > 1$: $f(x) = x^2 - 1$, $f'(x) = 2x$.
- $-1 < x < 1$: $f(x) = 1 - x^2$, $f'(x) = -2x$.
- Non-differentiable at $x = \pm 1$ (kinks).

> **Up to here**: Product = f'g+fg'. Quotient = (f'g−fg')/g^2. Chain = outside' × inside'. Implicit:

> differentiate both sides, attach dy/dx to y-terms, solve. Log-diff: take ln, simplify, differentiate, multiply by y. Parametric: dy/dx = (dy/dt)/(dx/dt).

Common Mistakes

Mistake 1: Forgetting the chain rule's inner derivative

Wrong: $\frac{d}{dx}\sin(x^2) = \cos(x^2)$. **Right**: $\cos(x^2) \cdot 2x$. The chain rule is NOT optional — every composition needs the inner derivative.

Mistake 2: Swapping the order in the quotient rule numerator

Wrong: $\frac{fg' - f'g}{g^2}$. **Right**: $\frac{f'g - fg'}{g^2}$. "Derivative of the TOP first." Check: $\frac{d}{dx}\frac{1}{x} = \frac{0 \cdot x - 1 \cdot 1}{x^2} = -\frac{1}{x^2} = -x^{-2}$. Power rule gives the same ✓.

Mistake 3: Dropping $\frac{dy}{dx}$ in implicit differentiation

Wrong: Differentiating y^2 as $2y$ with no $\frac{dy}{dx}$. **Right**: $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$. y is a function of x, not a variable.

Mistake 4: Using log-diff when a simpler rule works

Wrong: Log-diff on $y = x^2 \sin x$. **Right**: Product rule — faster and less error-prone. Log-diff is specifically for $f(x)^{g(x)}$ forms.

Mistake 5: Product rule on a constant times a function

Wrong: $\frac{d}{dx}(5\sin x) = 0 \cdot \sin x + 5 \cdot \cos x$. **Right**: Pull the constant out: $5 \cdot \frac{d}{dx}\sin x = 5\cos x$. The product rule works but is overkill.

What We Just Did

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(1) Product rule: (fg)' = f'g + fg'. Quotient: (f/g)' = (f'g−fg')/g².
    Decision: two factors→product. Fraction→quotient. Simple denominator→rewrite as x⁻ⁿ.

(2) Chain rule: (f(g(x)))' = f'(g(x))·g'(x). Peel from outside in.
    Multiple layers: write the derivative chain, multiply all layers.

(3) Implicit: differentiate both sides w.r.t. x. Attach dy/dx to every y-term.
    Collect dy/dx terms, factor, solve. Product rule applies to xy terms.

(4) Log-diff: y=f(x)^g(x) → ln y = g(x)·ln(f(x)). Differentiate, multiply by y.

(5) Parametric: dy/dx = (dy/dt)/(dx/dt). Inverse: (f⁻¹)'(y) = 1/f'(x).
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Practice 1

$f(x) = x^3 \cos x$. Run the product rule: $f = x^3$, $g = \cos x$. $f' =$?, $g' =$?, assemble.

→ Solutions: Solutions

Practice 2

$g(x) = \frac{e^x}{x^2+1}$. Run the quotient rule: top=e^x, bottom=$x^2 + 1$. Assemble numerator carefully.

→ Solutions: Solutions

Practice 3

$h(x) = \ln(\sin(x^2))$. Chain rule: how many layers? Peel from outside in. Multiply the chain.

→ Solutions: Solutions

Practice 4

$x^2 + xy + y^2 = 7$. Find $\frac{dy}{dx}$ at $(1, 2)$. Implicit: remember product rule on xy.

→ Solutions: Solutions

Practice 5

$y = (\cos x)^{\sin x}$. Log-diff: take $\ln$, simplify, differentiate, solve for y'.

→ Solutions: Solutions

Practice 6: Real Battle

$x = 2t - t^2$, $y = 3t^2 - t^3$. Find $\frac{dy}{dx}$ at $t = 1$ and the equation of the tangent line.

→ Solutions: Solutions

Basic Algebra Drill — Advanced Differentiation (10 Problems)

> Identify the structure. Apply the matching procedure.

D1. $\frac{d}{dx}(x^2 e^x)$. Product rule: $f = x^2$, $g = e^x$.

D2. $\frac{d}{dx}\left(\frac{\sin x}{x}\right)$. Quotient rule: top=$\sin x$, bottom=$x$.

D3. $\frac{d}{dx}((3x+2)^6)$. Chain rule: outer=$(\square)^6$, inner=$3x+2$.

D4. $\frac{d}{dx}\cos(5x)$. Chain rule: outer=$\cos(\square)$, inner=$5x$.

D5. $\frac{d}{dx}\ln(x^2+1)$. Chain rule: outer=$\ln(\square)$, inner=x^2+1.

D6. $\frac{d}{dx}\arcsin(3x)$. Inverse trig + chain: $\frac{1}{\sqrt{1-(3x)^2}} \cdot 3$.

D7. $\frac{d}{dx}\arctan(\sqrt{x})$. Inverse trig + chain: inner=$\sqrt{x}$.

D8. $\frac{d}{dx}(x^2 \sin x \cos x)$. Triple product OR group two factors.

D9. Find $\frac{dy}{dx}$ for $y^2 + x^2 y = 4x$. Implicit with product rule on $x^2 y$.

D10. $x = e^{2t}$, $y = \ln t$. Find $\frac{dy}{dx}$. Parametric: $dy/dt \div dx/dt$.

> Solutions: Solutions

Advanced Algebra Drill — Advanced Differentiation (10 Problems)

> Combine techniques. Prove identities. Handle edge cases.

A1. Differentiate x^x two ways: (a) log-diff, (b) $x^x = e^{x \ln x}$ + chain rule. Verify match.

A2. Find the 100th derivative of $\sin x$. Cycle length = 4. $100 \div 4 = 25$ remainder 0 → back to $\sin x$.

A3. $f(x) = \frac{x^2-1}{x^2+1}$. Find $f'(x)$ and simplify to a single fraction.

A4. $\sin(xy) = x + y$. Find $\frac{dy}{dx}$. Implicit: chain rule on $\sin(xy)$, product rule on xy.

A5. $f(x) = \arctan(\ln x) + \arcsin(e^{-x})$. Chain rule on both terms.

A6. Find tangent line to $x^3 + y^3 = 9xy$ at $(2, 4)$. Implicit diff, plug point.

A7. $f(x) = |x^3 - 3x|$. Find all x where f is NOT differentiable. (Solve $x^3 - 3x = 0$.)

A8. Prove $\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$. Implicit: $y = \arcsin x \to \sin y = x$. Differentiate.

A9. $f(x) = x^{x^x}$. Take $\ln$ twice: $\ln(\ln y) = x \ln x + \ln(\ln x)$? Actually: $\ln y = x^x \ln x$, then differentiate with product rule + chain.

A10. Cycloid: $x = t - \sin t$, $y = 1 - \cos t$. Find $\frac{dy}{dx}$ in terms of t. Simplify using half-angle identities. Find where the tangent is horizontal.

> Solutions: Solutions

Today's Procedure

When you see...	Do this...
$f(x) \cdot g(x)$	**Product rule**: $f'g + fg'$
$\frac{f(x)}{g(x)}$	**Quotient rule**: $(f'g - fg')/g^2$
$\frac{c}{g(x)}$ (constant top)	Rewrite as $c \cdot g(x)^{-1}$, use chain rule — faster
$f(g(x))$ (nested)	**Chain rule**: $f'(g(x)) \cdot g'(x)$. Peel outside→in
Multiple layers	Write the derivative chain. Multiply all layers
x and y mixed	**Implicit**: diff both sides, $y \to y'$, collect, solve
$f(x)^{g(x)}$	**Log-diff**: $\ln y = g \ln f$, differentiate, $\times y$
$x(t), y(t)$	**Parametric**: $dy/dx = (dy/dt)/(dx/dt)$
$\arcsin, \arccos, \arctan$	Dictionary + chain rule
Absolute value, piecewise	Break at zeros. Check each piece. Test differentiability at breaks

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Terminology

What we call it	Math term	Notation
differentiate a product	product rule	$(fg)' = f'g + fg'$
differentiate a quotient	quotient rule	$(f/g)' = (f'g - fg')/g^2$
differentiate nested functions	chain rule	$(f \circ g)' = (f' \circ g) \cdot g'$
peel from outside in	chain rule layers	multiply derivative chain
treat y as function of x	implicit differentiation	attach dy/dx to y-terms
take $\ln$ first	logarithmic differentiation	for $f(x)^{g(x)}$ forms
dy/dt over dx/dt	parametric derivative	$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$
derivative of inverse	inverse function theorem	$(f^{-1})'(y) = 1/f'(x)$